Estimating the dynamic correlation between an ENSO index and precipitation using the hydRopclim package in Rstudio

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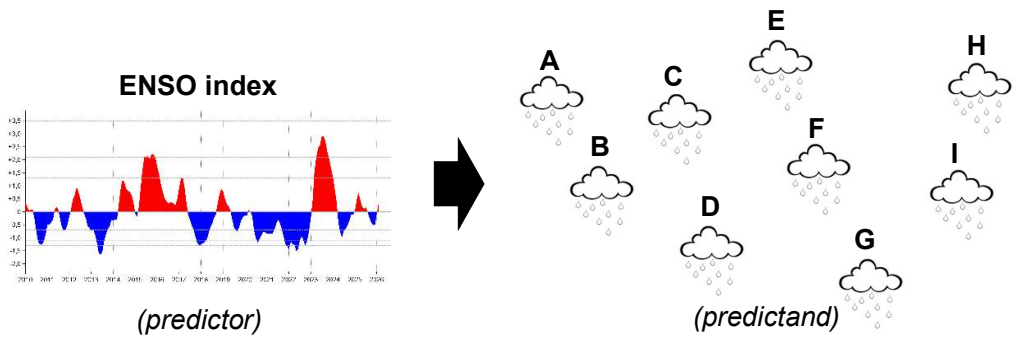
The following 10 time series represent a monthly ENSO (El Niño Southern Oscillation) index and the monthly Precipitation of 09 pluviometric stations ("A" to "I") in a study area.
Database: https://github.com/hydrocodes/hydRopclim/blob/main/tutorial/indexcorrl/data.csv



- Visualize the relationship between precipitation (as the predictand) for the season from January to March and the ENSO index (as the predictor) for the season from December to February. Use the Pearson correlation coefficient with a running window of 11 years, covering the period from 1960 to 2000.
- Identify and estimate the maximal and minimal correlation values, along with their corresponding years.
- Assess the trend in those correlations over the years.

Step 1: Install the ***hydRopclim*** package (Rau et al, 2023) on Rstudio, PositCloud or another user-defined IDE. First install the **remotes** package and then write in the code console.

```
remotes::install_github("hydrocodes/hydRopclim")
```

This step only executes once. If the package is already installed, start with step 2. Please check that your computer system requires to change regional settings to English format for matching the CSV format (decimals with period, no commas to separate thousands).

Step 2: *hydRopclim* works with *csv* files. Create a file **input.csv** with the initial data with the following format including the names of the headers: *Date, Index*, and the names of the stations (A to I). The information must start in January and ending in December with enough number of years to do a running correlation with a defined window. Date format is %b-%Y (months with the three letters and the first in majuscule and the year with complete 4 numbers).

Date	Index	A	B	C	D	E	F	G	H	I
Jan-1960	0.3047	0.2	13.9	158.2	8.4	140.9	193	7.1	117.8	175.6
Feb-1960	0.2884	0.5	22.4	116.1	1.1	27.6	61	2.2	75.6	150.4
Mar-1960	-0.0231	3.9	18.3	129.7	1	121.6	63.5	0.6	46.9	158.7
..										
Dec-2000	-1.2747	0.2	2.9	70.2	0.8	93.9	69.6	0.9	27.4	92.8

Step 3: Read the created file input.csv with the object name "**data**". Extract the index values with the variable name **"index"** and extract the precipitation values from the 9 stations with the variable name **"variables"**. Define the running or sliding window in a number of years with the variable name **"rwin"**.

```
data <- read.csv("C:/.../data.csv")
index <- data$Index
variables <- data[ ,3:ncol(data)]
rwin <- 11
```

Step 4: Define the path and name of three output files. They will store the indexes of the ENSO and the precipitation, all processed for the seasonal period to consider in the Step 5, the running correlations along the years, and a special format for plotting in external softwares.

```
output1 <- "C:/.../indexes.csv"
output2 <- "C:/.../runcorr.csv"
output3 <- "C:/.../runcorr_format.csv"
```

Step 5: Load the libraries required as complements of hydRopclim, which allow to arrange and color the plots appropriately.

```
library(hydRopclim)
library(reshape2)
library(ggplot2)
library(wesanderson)
library(cowplot)
```

Step 6: The most important step of the analysis is to define the seasonal period to obtain the interannual indices. Define the start and the end of the month for both the predictor and the predictand in a format of the number of the month, following the logic of the next table.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Ago	Sep	Oct	Nov	Dec
Number	1	2	3	4	5	6	7	8	9	10	11	12

For the ENSO index, being just one column in the data, we use an average function called ***seasavg***. For example, to convert the month to interannual time series starting from December (month 12) to February for the predictor. It accounts a total of 3 months (December, January and February, as a ***win***=3). Run the next code:

```
p1 <- seasavg(p=index, start=12, win=3) # e.g. DJF
```

For the Precipitation data, being many columns in the data for each station, we use a sum function called ***seassum*** (for other variables as temperature or streamflow, it can be used the function called ***seasavg2***). For example, to convert the month to interannual time series starting from January (month 1) to March for the predictand. It accounts a total of 3 months (January, February and March, as a ***win***=3). Run the next code:

```
p2 <- seassum(p=variables, start=1, win=3) #e.g. JFM
```

Step 7: Run the ***indexcorrl*** function

```
indexcorrl(index.seas=p1, variable.seas=p2, rwin=11)
```

Check the results in the console: In the first level of the list [[1]] you will find the seasonalized ENSO index (from December to February).

```
[[1]]
 [1] -1.35456667 -0.70133333  1.57796667 -1.74323333 -0.57446667 -1.48970000
 [7]  0.55470000  0.52120000 -0.06966667  0.43503333 -0.25263333 -0.01243333
[13]  2.02596667 -0.48006667 -0.89820000 -0.48436667  1.09510000  ...
```

In the second level [[2]] you will find the seasonalized precipitation indices for each station (from January to March).

```
[[2]]
       [,1]   [,2]   [,3] [,4]  [,5]  [,6] [,7]  [,8]  [,9]
 [1,]  67.4  289.6  629.0  6.5 297.6 278.4  3.5 191.0 383.2
 [2,] 149.7  451.0  651.0 20.1 410.7 422.7 22.4 442.6 530.7
 [3,]  48.4  342.4  591.5 17.0 327.0 298.2 27.4 284.7 ...
```

In the third level [[3]]: the running correlation for the 11-years window used for each of the stations with their names.

```
[[3]]
              A           B            C           D             E           F
1966 -0.2001438  0.05725018 -0.389099464  0.01520844 -1.077253e-01 -0.29587817
1967 -0.1432130  0.12313937 -0.342325365 -0.08948864 -9.389811e-02 -0.31562369
1968  0.5684569  0.62371703  0.077714033  0.15604731  2.802474e-01 ...
```

In the fourth level [[4]]: the slope of the relationship of the running correlation over time (+ and - values) for each station (*X1.ncol.rcorr* are inverted to the name of stations, 2 is H).

```
[[4]]
     rev.slope. X1.ncol.rcorr.
1  0.0005565879             1
2 -0.0038796600             2
3 -0.0114415214             3 ...
```

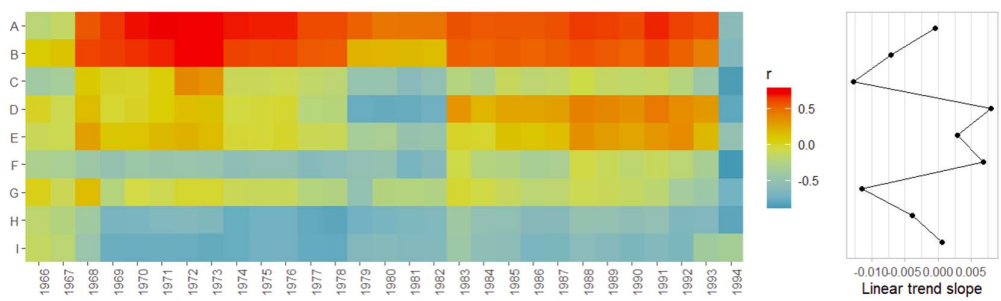


Fig 1. hydRopclim output for the function ***indexcorrl***. The left figure illustrates the dynamic correlation over time, highlighting the strength and weakness of both direct and inverse relationships between ENSO and precipitation. The right figure displays the trend in these correlations.

Answering the three initial questions of this applied case:

- Figure 1 highlights direct (red) and inverse (blue) relationships. Red shows Pearson coefficients above 0.5, with strong periods at stations A and B. Blue indicates coefficients below -0.5, mainly at stations H and I.
- Extreme correlation values are detailed in the console/output files. The highest value (0.79) occurs in 1972 at station B, followed by 0.77 at station A. The lowest value (-0.895) appears in 1994 at station F. Weak values (-0.14 and -0.16) are observed at stations G and I.
- Trend analysis reveals a peak positive value of 0.00799 at station D and a negative value of -0.0128 at station C.

Reference:
Rau P, Castillon F, Bourrel L. 2023. A tool in R for easy hydroclimatic calculations. In: Recent Research on Hydrogeology, Geoecology and Atmospheric Sciences. MedGU 2021. Advances in Science, Technology and Innovation. 13-16. https://doi.org/10.1007/978-3-031-43169-2_3

Github repository: https://github.com/hydrocodes/hydRopclim